



DISSERTATION PRESENTATION



In silico Evaluation of Phytochemicals for Combating Antibiotic Resistance in *Klebsiella pneumoniae*

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The Antibiotic Resistance Crisis

- Antibiotic Resistance (ABR) is a critical global public health threat, undermining the effectiveness of essential medicines.
- Driven by Misuse, overuse of antibiotics and Incomplete treatment courses
- **GLASS 2022 report** highlights high resistance in Gram-negative bacteria:
 - **42%** resistance in *E. coli* to third-generation cephalosporins:
 - Approximately **1 in 5** Urinary Tract Infections (UTIs) are resistant to common antibiotics.
- Similar resistance trends are observed in ***Klebsiella pneumoniae***, a critical opportunistic pathogen.
- **OECD projection:** Resistance to last-resort drugs may **double by 2035**.

Klebsiella pneumoniae

Classification

- Gram-negative, facultative anaerobe
- Family: Enterobacteriaceae

Morphology

- Rod-shaped bacterium
- Thick polysaccharide capsule
- Non-motile

Habitat

- Normal flora of intestinal tract
- Found in soil, water, and hospital environments

Diseases

- Pneumonia, Urinary tract infections
- Septicemia (bloodstream infections)
- Wound infections

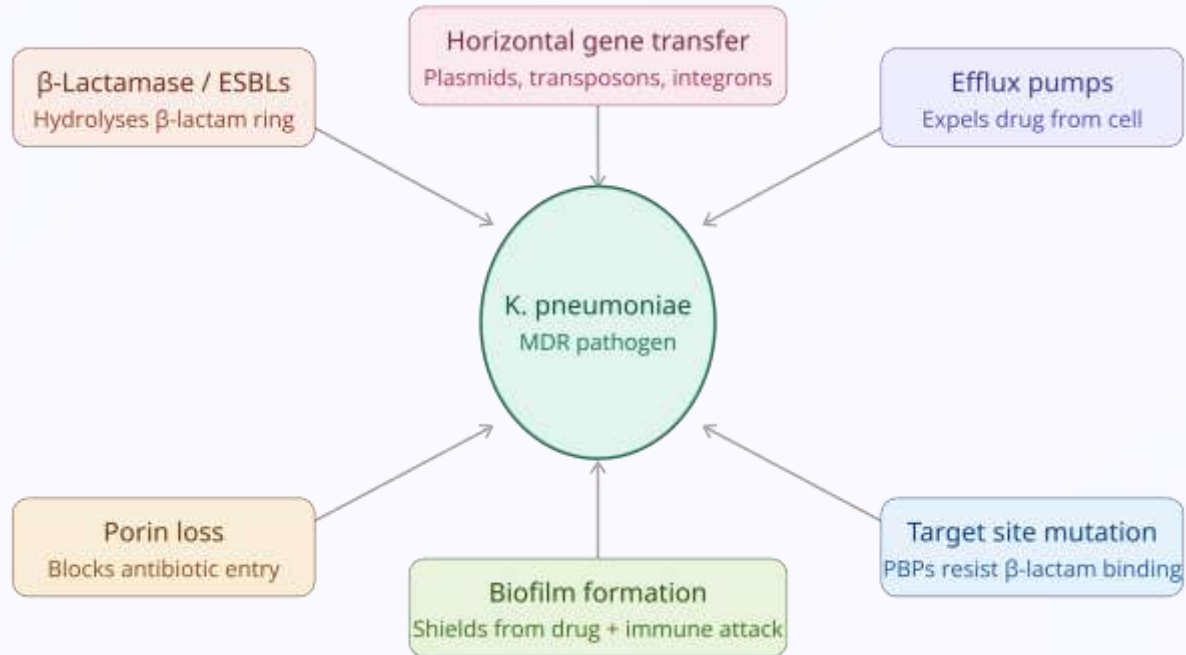
Virulence Factors

- Capsule - prevents phagocytosis
- Lipopolysaccharide (LPS) - endotoxin activity
- Siderophores - iron acquisition

Clinical Significance

- Major cause of hospital-acquired infections (HAIs)
- Exhibits multidrug resistance (MDR)
- Includes Beta lactamase production

Mechanisms of Antibiotic Resistance in *Klebsiella pneumoniae*



Penicillin-Binding Proteins (PBPs) as Target proteins

Penicillin-Binding Proteins (PBPs) are crucial membrane-associated enzymes in *Klebsiella pneumoniae*, integral to bacterial survival. They catalyze the final stages of peptidoglycan synthesis, forming the bacterial cell wall and are the primary targets of β -lactam antibiotics.

Functions of Selected PBPs

PBP	Function
PBP1A	Peptidoglycan polymerization and cell wall elongation
PBP1B	Transglycosylation and transpeptidation
PBP ₃	Septum formation during bacterial cell division

Conserved Active-Site Motifs

These motifs are critical for PBP function:

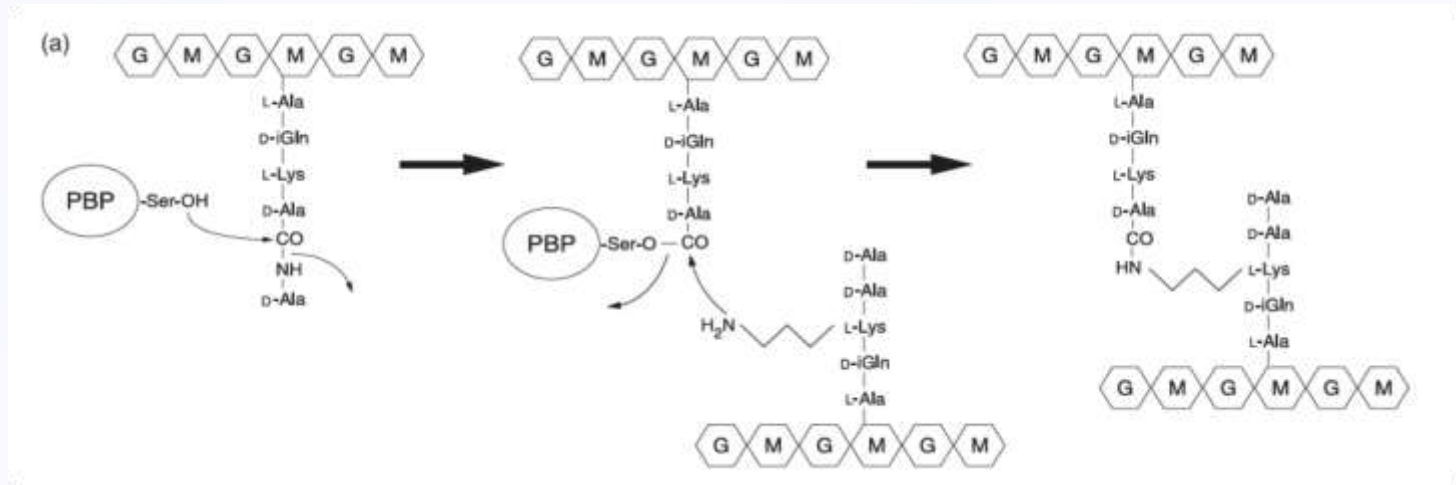
- **SXXK** - catalytic serine motif
- **SXN** - substrate stabilization
- **KTG** - active-site organization

Why PBPs Were Selected

- Essential for bacterial survival
- Conserved catalytic motifs
- Clinically validated antibiotic targets
- Associated with β -lactam resistance mechanisms

β -Lactam Antibiotics and Mechanism of Action

Bacterial cell wall synthesis. Adapted from *Zapun et al., 2008*.



β -Lactam Antibiotics

- Major class of antibacterial agents targeting bacterial cell wall synthesis
- Characterized by the presence of a β -lactam ring

Examples: Penicillin, Amoxicillin, Dicloxacillin, Cefixime, Carbapenems

Phytochemicals as Antibacterial Agents

Phytochemicals are naturally occurring bioactive compounds derived from plants, offering a promising avenue to combat multidrug-resistant (MDR) bacterial infections.

Key Phytochemical Classes with Antimicrobial Activity:

- **Terpenoids:** Known for membrane disruption and enzyme inhibition.
- **Flavonoids:** Interfere with nucleic acid synthesis and cell wall integrity.
- **Alkaloids:** Inhibit bacterial DNA/RNA synthesis and protein function.
- **Sesquiterpene Lactones:** Highly potent with diverse mechanisms, including oxidative stress.

These compounds exhibit structural diversity, multi-target activity, and a lower probability of resistance development compared to conventional antibiotics.

Study Relevance:

Sesquiterpene lactones, in particular, were selected for this study due to their reported potent antimicrobial potential and demonstrated capability for protein interaction.



Terpenoids

Found in rosemary, mint.



Flavonoids

Abundant in citrus, berries.



Alkaloids

Derived from poppy, coffee.



Sesquiterpene Lactones

Present in chicory, dandelion.

Study Rationale



Problem: Antimicrobial Resistance

Rapid emergence of MDR *K. pneumoniae* limits treatment options, especially against β -lactam and carbapenem resistance.



Need: Novel Agents

Urgent demand for new antibacterial compounds; phytochemicals offer diverse structures with reported antimicrobial properties.



Target: Essential PBPs

PBPs (1A, 1B, 3) are crucial enzymes for bacterial cell wall synthesis and clinically validated β -lactam drug targets.



Approach: In silico Screening

Virtual high-throughput screening efficiently identifies potential PBP inhibitors from large phytochemical libraries.



Significance: Alternative Strategies

Identify non- β -lactam phytochemicals targeting PBPs to combat resistant *K. pneumoniae* infections.

Objectives

1 Identify PBP Targets

Characterize and select relevant Penicillin-Binding Protein structures from *K. pneumoniae*

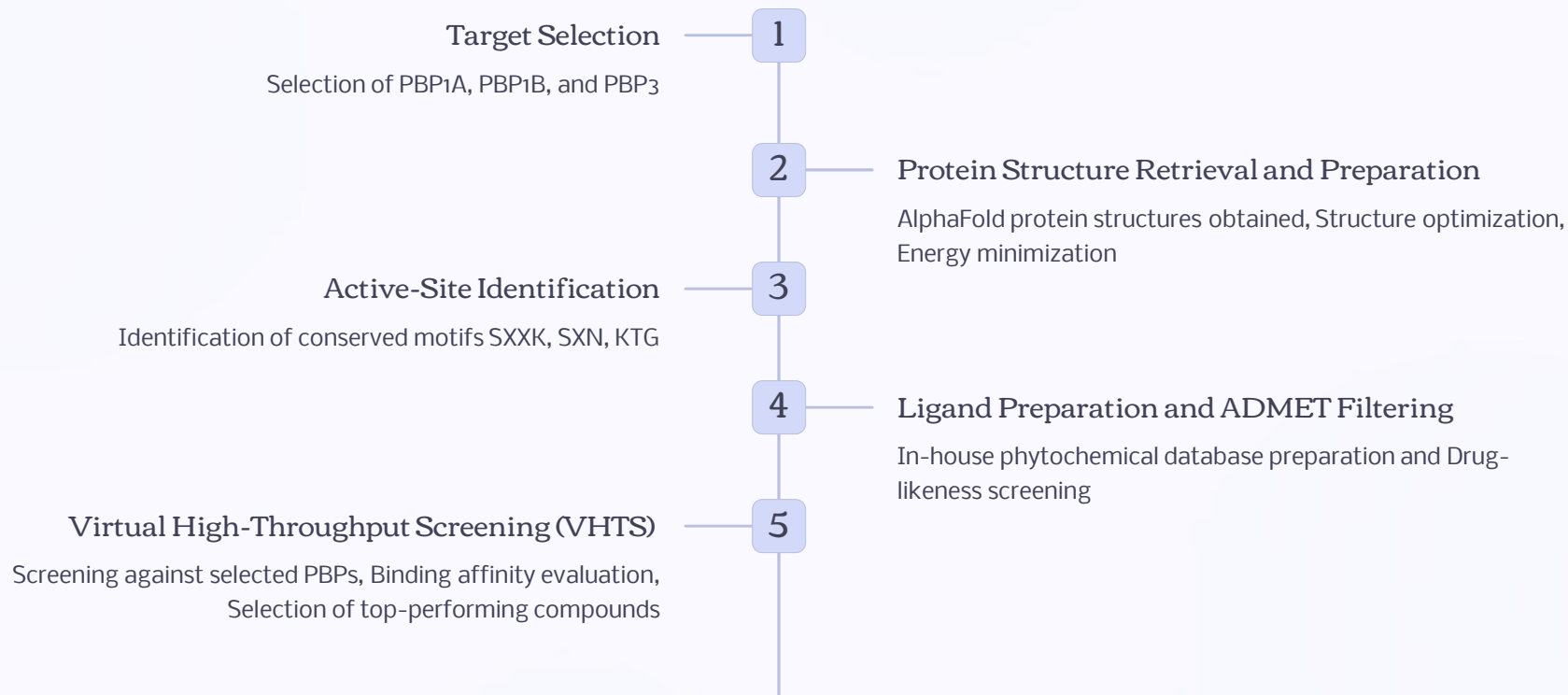
2 Protein and Ligand Preparation

Structure optimization and energy minimization, to ensure suitability for virtual high-throughput screening (VHTS)

3 Virtual high-throughput screening (VHTS)

VHTS of an in-house phytochemical database against the selected targets

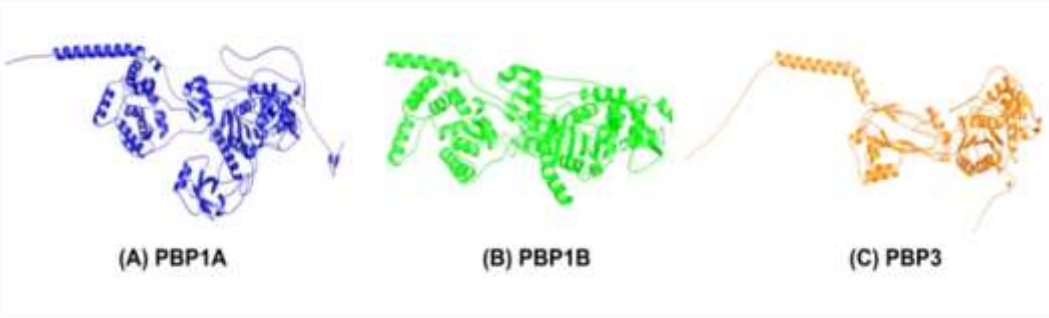
Computational Workflow



Protein Target Selection

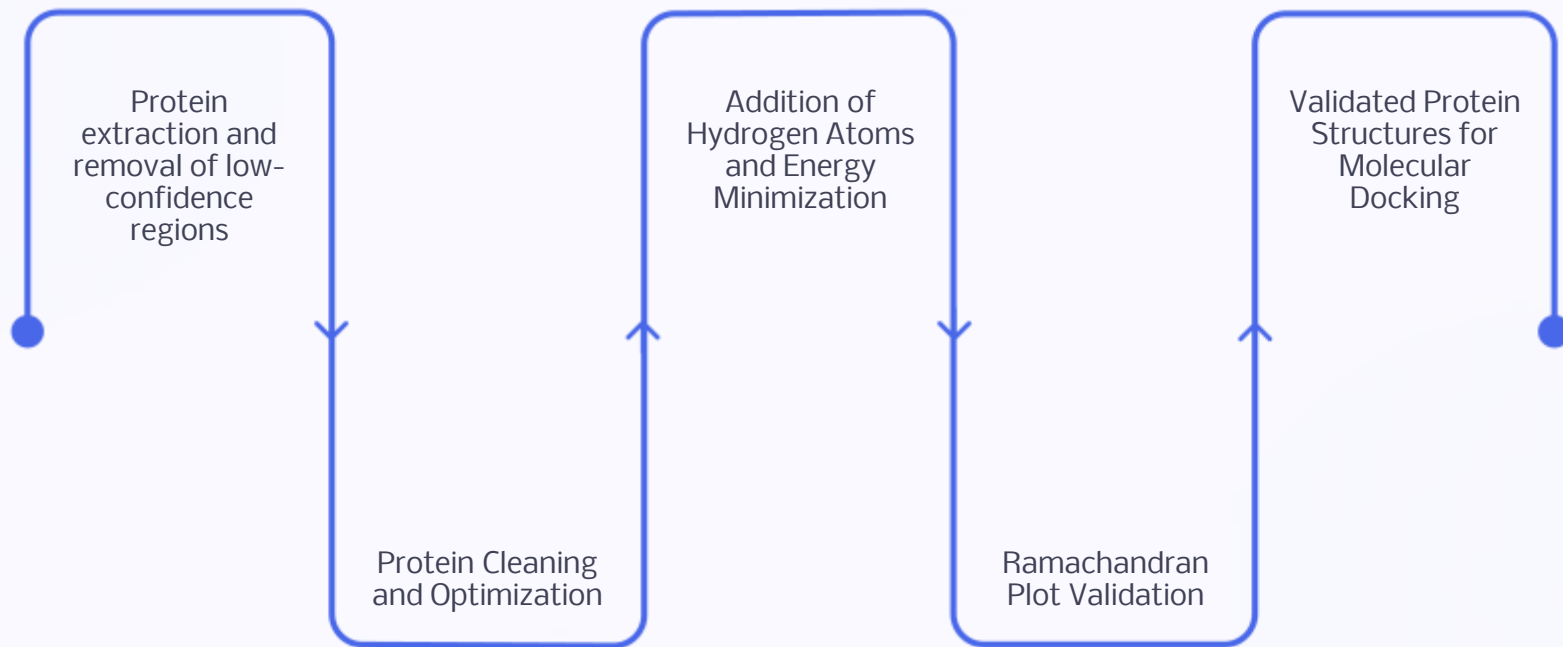
The shown structures of PBP1A, PBP1B, and PBP3 were obtained from the AlphaFold Protein Structure Database.

Protein Name	UniProt ID	Length (aa)	Function
PBP1A	AoAAW9Vo26	862	Involved in peptidoglycan synthesis and cell wall elongation
PBP1B	AoA486VT81	845	Plays a role in cell wall biosynthesis and maintenance
PBP3	AoAoW8AVJ6	588	Responsible for septum formation during cell division



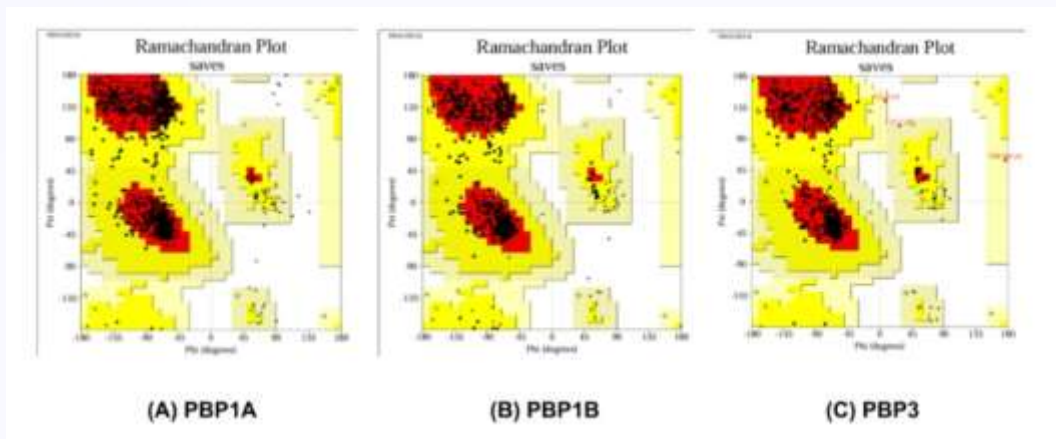
Protein Structure Preparation and Validation

Protein structure preparation and validation is done by using **UCSF Chimera**.



Ramachandran plot Analysis

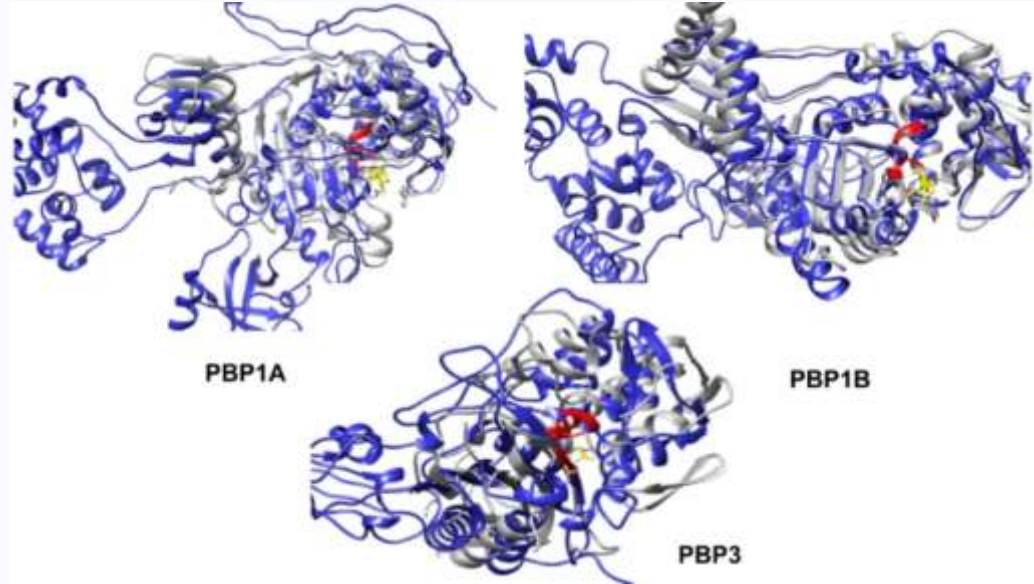
Protein	Favoured Regions (%)	Allowed Regions (%)	Generously Allowed (%)	Disallowed Regions (%)
PBP1A	90.6	9.4	0	0
PBP1B	90	10	0	0
PBP3	91.2	8.2	0.6	0



Ramachandran plot analysis was performed to validate the stereochemical quality of the protein structures. Most residues were observed in favoured regions, with no residues in disallowed regions, indicating reliable structures suitable for docking studies

Active-Site Identification and Validation

- Conserved catalytic motifs (SXXK, SXN, and KTG) were identified from published literature and manually verified through sequence analysis.
- Structural superimposition with reference *Streptococcus pneumoniae* crystal structures was performed to validate active-site localization.
- Co-crystallized ligands confirmed binding within the identified catalytic regions.



 The modeled protein is shown in blue, while the reference proteins are represented in gray. The co-crystallized ligands are depicted in yellow, indicating the active site region. Conserved active site residues are highlighted in red, demonstrating structural alignment and conservation of the binding pocket across all structures.

In-house Phytochemical Database

Our study employed a rigorous computational workflow to identify potential inhibitors of *Klebsiella pneumoniae* PBPs from a vast array of phytochemicals.

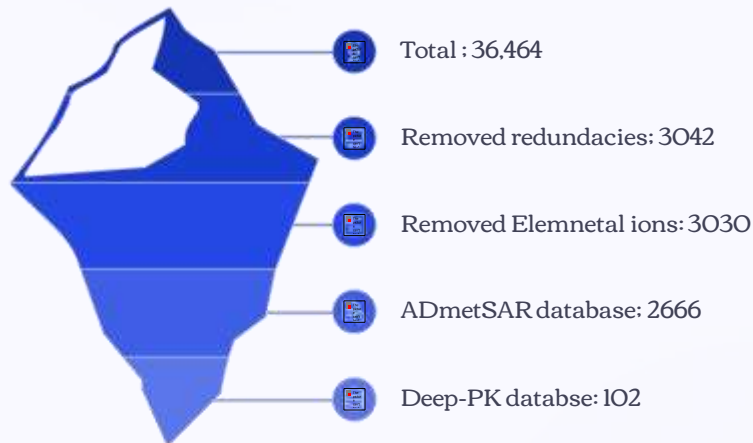
Database Sources

- Dr. Duke's Phytochemical and Ethnobotanical Database
- Traditional Chinese Medicine Systems Pharmacology (TCMSP)
- Herbs Database
- SymMap Database

✓ 102 Phytochemicals selected

☐ Screening Criteria

- Drug-likeness
- ADMET profiling
- Bioavailability
- Toxicity prediction
- Intestinal absorption
- Blood-brain barrier penetration
- CYP inhibition potential



Docking and Virtual High-Throughput Screening

Batch Virtual High-Throughput Screening (VHTS)

- A library of 102 phytochemicals was screened using RACCOON v1.0 and AutoDock 4.2.
- Binding energies (ΔG , kcal/mol) were calculated against all three PBPs.
- Compounds with stronger binding than antibiotic controls were shortlisted for detailed analysis.
- Ligand 101 demonstrated the strongest overall binding affinity across all targets.



Docking of Standard Antibiotics

Reference Antibiotics

- Dicloxacillin
- Amoxicillin
- Cefixime

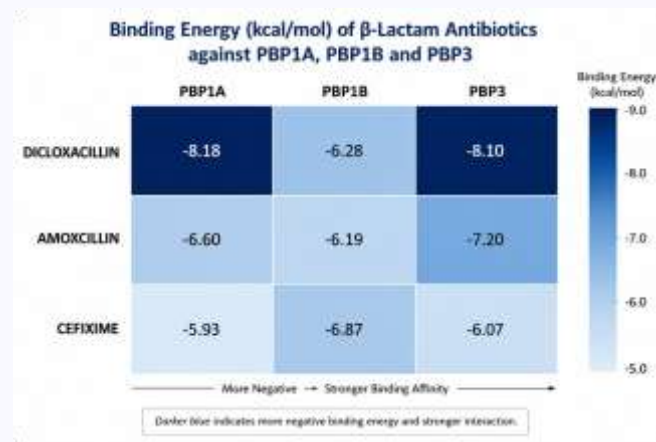
Methodology

Docking was performed using AutoDock 4.2 to establish a validated baseline for comparison with phytochemicals.

Key Findings

- All antibiotics exhibited negative binding energies across all three PBPs, confirming interaction with active sites.
- Dicloxacillin demonstrated the strongest average binding affinity among the tested antibiotics.
- Dicloxacillin was selected as the primary benchmark comparator for lead phytochemical compounds.

Binding energy (kcal/mol) of reference β -lactam antibiotics against *K. pneumoniae* penicillin-binding proteins.



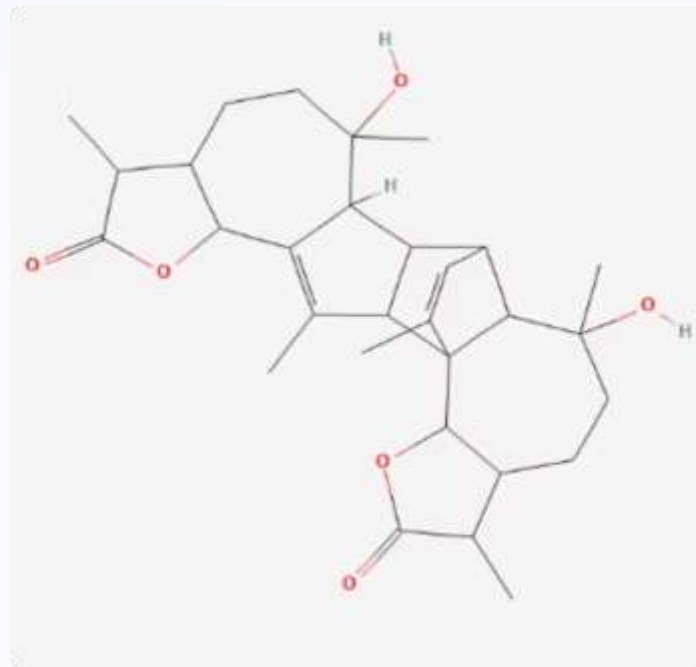
Lead Compound Identification: Absinthin

Lead Compound: Absinthin (Ligand 101)

- Identified as the top-performing phytochemical across all PBPs
- PubChem CID: **348165**
- Molecular Formula: **C₃₀H₄₀O₆**
- Molecular Weight: **496.64 g/mol**

Structural Classification

- Dimeric guaianolide sesquiterpene lactone
- Derived from *Artemisia absinthium* (Wormwood)
- Contains:
 - polycyclic scaffold
 - multiple lactone rings
 - diverse functional groups



2D structure of Absinthin

ADMET Profile of Absinthin

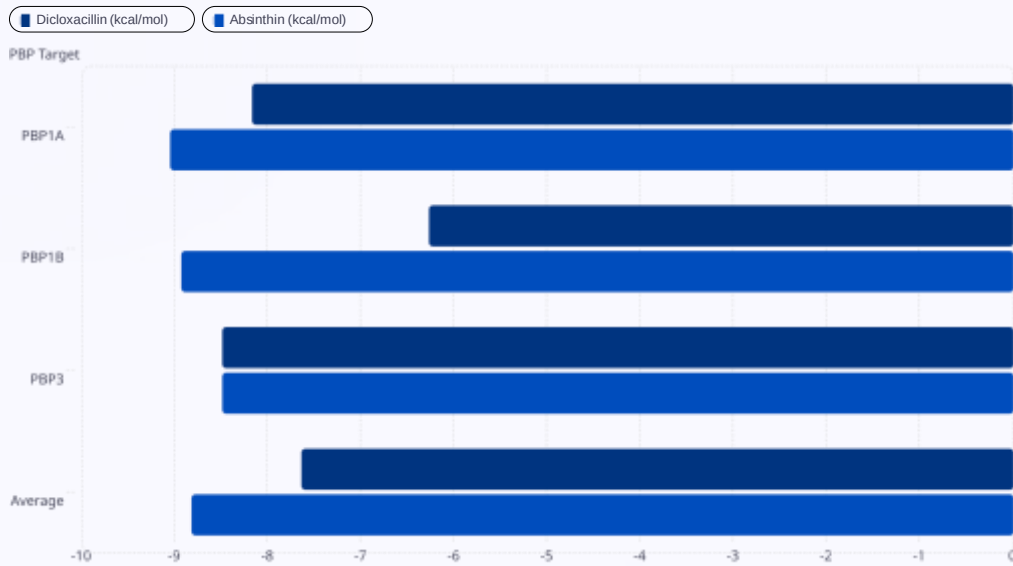
Parameter	Prediction	Confidence
Oral bioavailability	Bioavailable	High
Intestinal absorption	Absorbed	High
BBB penetration	Non-penetrable	High
P-gp inhibitor	Non-inhibitor	High
CYP inhibition	Non-inhibitor	High
AMES toxicity	Safe	Medium
Carcinogenicity	Safe	Low
DILI	Safe	Low

Key Findings

- Favorable oral absorption and bioavailability predicted
- Low risk of CYP-mediated drug interactions
- Non-penetrable to BBB, suggesting reduced CNS toxicity risk
- Predicted non-mutagenic and non-carcinogenic profile

Binding Energy Comparison

Comparative binding energies of Dicloxacillin and Absinthin against *K. pneumoniae* PBPs.



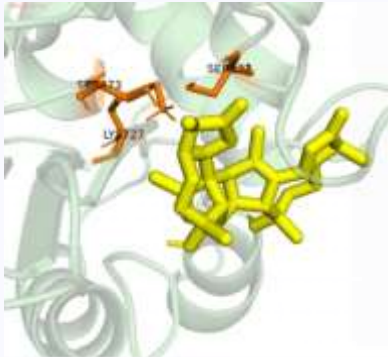
Key Findings

- Absinthin outperformed Dicloxacillin against PBP1A and PBP1B, showing generally stronger (more negative) binding energies.
- The **largest improvement** was observed against PBP1B, with Absinthin binding significantly stronger than Dicloxacillin (+2.66 kcal/mol difference in binding affinity).
- For PBP3, Absinthin showed **comparable binding** affinity to Dicloxacillin.

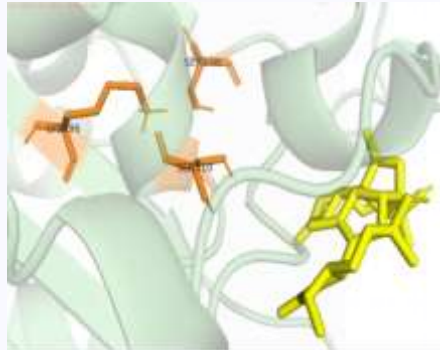
Binding Site Analysis and Interaction Mode

- Docking poses of Absinthin were visualized using **PyMOL** against PBP1A, PBP1B, PBP3
- Absinthin consistently localized at the **entrance of the transpeptidase active-site pocket** across all three targets.
- Binding occurred near conserved catalytic motifs **SXXK, SXN, KTG**
- The compound interacted primarily with **entrance-lining residues** rather than directly contacting catalytic residues.

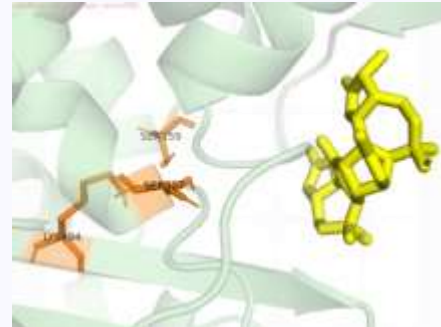
Three-dimensional docking poses of Absinthin (yellow sticks) at the entrance of the transpeptidase active site of *K. pneumoniae* PBP1a visualized in PyMOL. Active site residues are shown as orange sticks. The protein backbone is rendered as a light green ribbon.



PBP1A



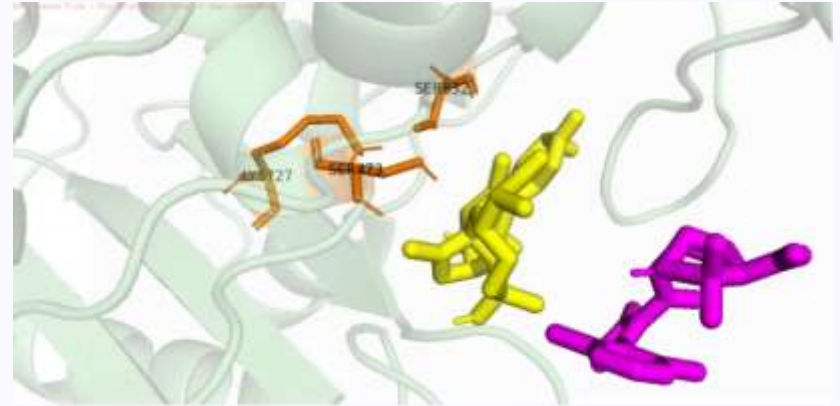
PBP1B



PBP3

Superimposed Docking Analysis

- Superimposition was performed between **Absinthin** and **Dicloxacillin** within the active sites of PBP1A, PBP1B, PBP3
- Both ligands occupied the **entrance region** of the transpeptidase active-site pocket.
- Distinct and non-overlapping binding orientations were observed for the two compounds.
- Dicloxacillin functions through **β -lactam-mediated catalytic serine acylation**, whereas absinthin demonstrated stable **non-covalent entrance-site binding**.



Superimposed docking poses of Absinthin (yellow sticks) and Dicloxacillin (magenta sticks) at the entrance of the PBP1a and transpeptidase active sites of *K. pneumoniae*, visualized in PyMOL. Active site residues are displayed as orange sticks. The protein backbone is rendered as a light green ribbon

Protein-Ligand Interaction Analysis

Parameter	PBP1A	PBP1B	PBP3
Catalytic Motifs	SXXK, SXN, KTG	SXXK, SXN, KTG	SXXK, SXN, KTG
Binding Position	Entrance	Entrance	Entrance
H-bonds	3	1	2
Interaction Type	H-bond + Hydrophobic	Mainly Hydrophobic	H-bond + Hydrophobic
Binding Energy	−9.06	−8.94	−8.50

- 2-D interaction analysis was performed using **LigPlot+** to evaluate Absinthin binding interactions with all three proteins.
- Absinthin formed stable **hydrogen bonds** and extensive **hydrophobic interactions** at the active-site entrance of all three PBPs.

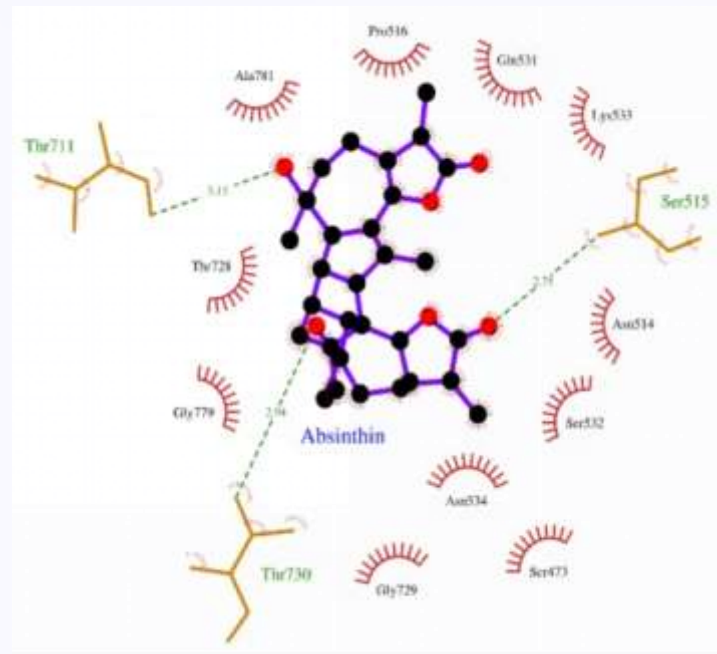
LigPlot+ Interaction Analysis: PBP1A

PBP1A Interaction Analysis

- Absinthin formed **3 hydrogen bonds** with:
 - Thr711
 - Ser515
 - Thr728
- Extensive hydrophobic interactions stabilized the ligand at the active-site entrance.
- Strongest binding affinity observed: **-9.06 kcal/mol**

Key Observation

- Combined hydrogen bonding and hydrophobic enclosure contributed to highly stable binding



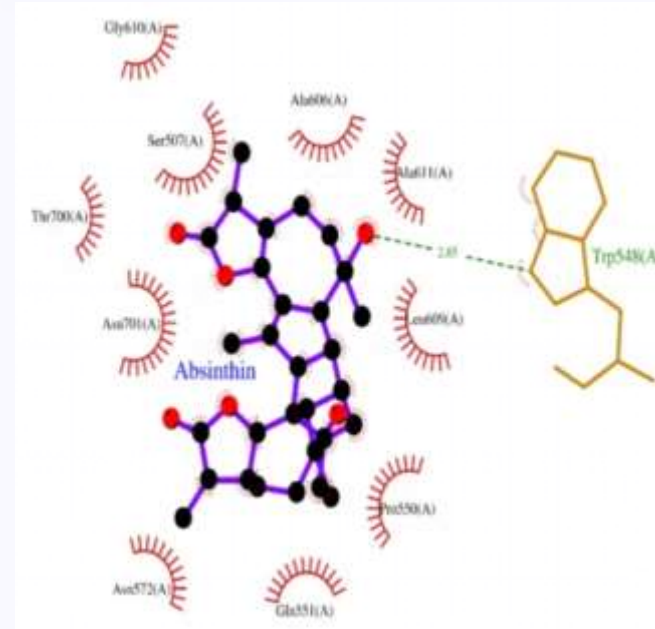
LigPlot+ Interaction Analysis: PBP1B

PBP1B Interaction Analysis

- Absinthin formed **1 hydrogen bond** with:
 - Trp548
- Multiple hydrophobic contacts contributed to interaction stability.
- Largest improvement over Dicloxacillin observed: **+2.66 kcal/mol**

Key Observation

- Hydrophobic interactions appeared to play a major role in stabilization within PBP1B.



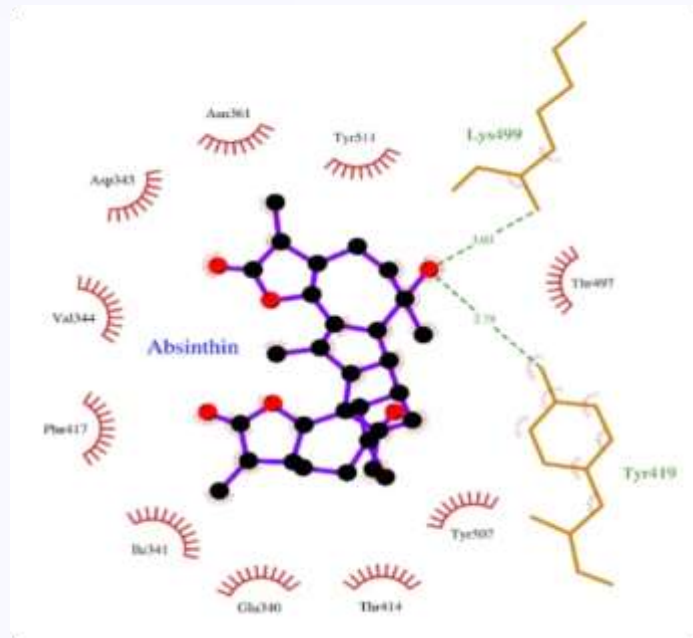
LigPlot+ Interaction Analysis: PBP3

PBP3 Interaction Analysis

- Absinthin formed 2 hydrogen bonds with:
 - Lys499
 - Tyr419
- Stable hydrophobic interactions observed with surrounding entrance residues.
- Binding affinity recorded: **-8.50 kcal/mol**

Key Observation

- Consistent entrance-binding geometry and non-covalent interaction profile were maintained in PBP3.



Literature Support and Biological Plausibility

Literature Evidence

- Previous Studies:
 - Sesquiterpene lactones exhibit antimicrobial activity (Moujir et al., 2020)
 - α,β -unsaturated γ -lactone groups act as Michael acceptors (Rüngeler et al., 1999)
 - Phytochemicals can show binding affinities comparable to β -lactam antibiotics (El-Gendy et al., 2022)

Present Study Findings

- Absinthin localized at the entrance of the PBP active-site pocket
- Stable interaction supported by:
 - hydrogen bonding
 - hydrophobic interactions
- Suggested mechanism:
 - steric occlusion / active-site blocking

Key **Significance:** Supports the therapeutic potential of Absinthin as a non- β -lactam PBP inhibitor.

Key Findings

Summarizing the major outcomes of the computational screening and docking study against *Klebsiella pneumoniae* PBPs.



VALIDATED TARGETS

PBP1A, PBP1B, PBP3 selected with **90%** residues in favored Ramachandran regions.

Highlight: Reliable protein models for docking.



PHYTOCHEMICAL SCREENING

36,464 initial compounds screened, yielding **102** ADMET-filtered phytochemicals.

Highlight: High-quality screening library.



LEAD COMPOUND

Absinthin (Ligand 101) identified as a dimeric guaianolide sesquiterpene lactone.

Highlight: Top-performing phytochemical.



BINDING AFFINITY

Absinthin showed strong binding: PBP1A (**-9.06 kcal/mol**), PBP1B (**-8.94 kcal/mol**), PBP3 (**-8.50 kcal/mol**).

Highlight: Comparable or superior affinity to Dicloxacillin.



INTERACTION MECHANISM

Entrance-site localization with non-covalent interactions, proposing a steric occlusion mechanism.

Highlight: Distinct from classical β -lactam inhibition.



ADMET PROFILE

Predicted oral bioavailability, non-BBB penetrable, and low predicted toxicity.

Highlight: Favorable pharmacokinetic profile.

Future Directions



Molecular Dynamics (MD) Simulations

Evaluate the stability and dynamics of Absinthin-PBP complexes over time.



In Vitro Assays

Conduct antibacterial and enzyme inhibition assays to validate computational findings experimentally.



MIC Evaluation

Determine minimum inhibitory concentrations (MICs) against multidrug-resistant *Klebsiella pneumoniae* isolates.



Structural Optimization

Design and synthesize Absinthin derivatives to improve potency, selectivity, and pharmacokinetic properties.



SAR Studies

Perform structure - activity relationship studies to identify key pharmacophores for PBP inhibition.



Long-Term Goal

Ultimately, this research aims to facilitate the development of novel non- β -lactam antibacterial agents targeting PBPs to combat antibiotic resistance.

Conclusion

Summarizing the major outcomes and scientific significance of the study.



Lead Compound Identified

Absinthin (Ligand 101) emerged as the **top-performing phytochemical** against PBP1A, PBP1B, and PBP3.



Favorable Profile

The compound demonstrated **strong and consistent binding affinity**, stable entrance-site interactions, and **favorable ADMET properties**.



Mechanism of Action

Interaction analysis supported a possible **steric occlusion mechanism**, distinct from classical β -lactam antibiotics.



Therapeutic Potential

Findings support the therapeutic potential of **phytochemical-based, non- β -lactam PBP inhibitors** against multidrug-resistant *Klebsiella pneumoniae*.

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Key scientific and methodological sources underpinning this research.

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Thank You for Your Attention

Questions and Discussion